CDS TapeLab – User Manual

1. What is CDS TapeLab

CDS TapeLab is a Windows program for measuring cassette decks. Together with an ASIO sound card (up to 192 kHz; this manual uses the Behringer UMC202HD as an example) it replaces several separate instruments. After starting the program you choose one of four measurements:

Measurement	What it measures	What you need
Wow, flutter & speed	speed deviation, wow & flutter (JIS and DIN), spectrum	test cassette with a 3000 or 3150 Hz tone
Playback frequency response	playback frequency response, 16 Hz – 25 kHz, 1/3-octave bands	test cassette with a sweep
Recording frequency response	recording frequency response at levels −20, −10, 0 and +6 dB	blank cassette, 3-head deck with tape monitor
Head azimuth	L/R phase difference of a 10 / 12.5 / 15 kHz test tone; live adjustment of the playback azimuth	test tape with an azimuth tone

The interface is available in Slovenian and English (switch at the top right). Start the program with the CDS TapeLab icon on the desktop.

2. Installation and licence

The program is fully usable in demo mode: all measurements work, but export of images and CSV is disabled until you enter a licence.

System requirements

The minimum screen resolution at which the program works normally is 1280 × 960 pixels. At a lower resolution parts of the window are not fully visible or legible. The program is intended for 64-bit Windows.

Installation

1. Run the file `CDSTapeLab-Setup-0.3.0.exe` and follow the wizard (Next → Install).
2. A **CDS TapeLab** icon appears on the desktop. Double-click it.
3. You do not need to uninstall the previous version: the new one installs over the old one.

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The ASIO driver of your sound card must be installed in Windows (for example, Behringer offers the driver for the UMC202HD free of charge). The program is not tied to one sound card; the UMC202HD is only the example used in this manual. Without it the program reports "No ASIO driver found."

Licence

1. On the start screen click the **Licence** button.
2. The window shows the **Computer ID**. It is tied to your computer.
3. Enter the **purchase code** or **licence key** you received in the field.
4. Click **Activate**. The computer must be connected to the internet to activate a code.
5. When the message "Licence activated. Thank you!" appears, export (image, CSV, report) is unlocked.

A code works on one computer only. For another computer you need a new code. Messages you may see: "This code is already activated on another computer.", "Invalid code or key.", "Could not reach the server. Check your internet and try again." (check your internet connection).

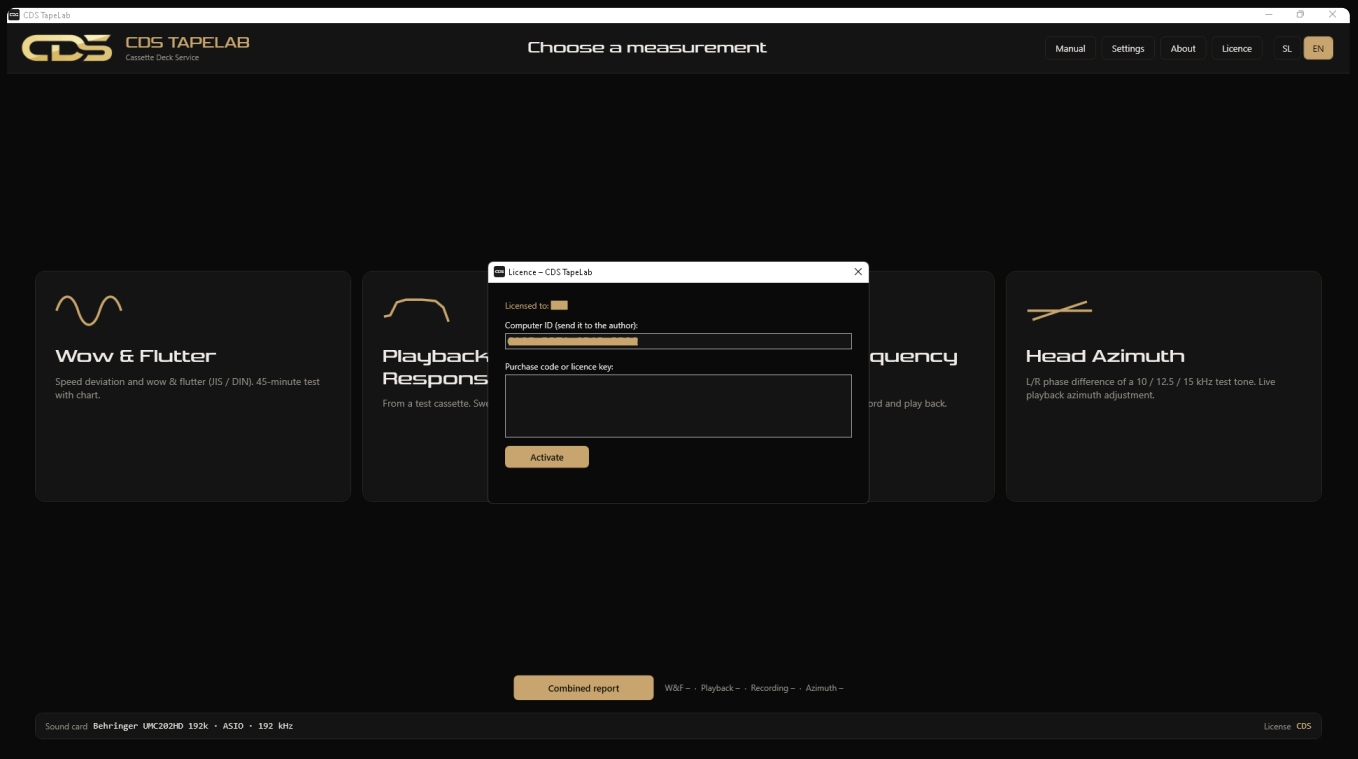


Figure: the Licence window; enter the purchase code or licence key in the field below.

Settings and About

Besides the **Licence** button, the top bar of the main window has the **Settings** and **About** buttons.

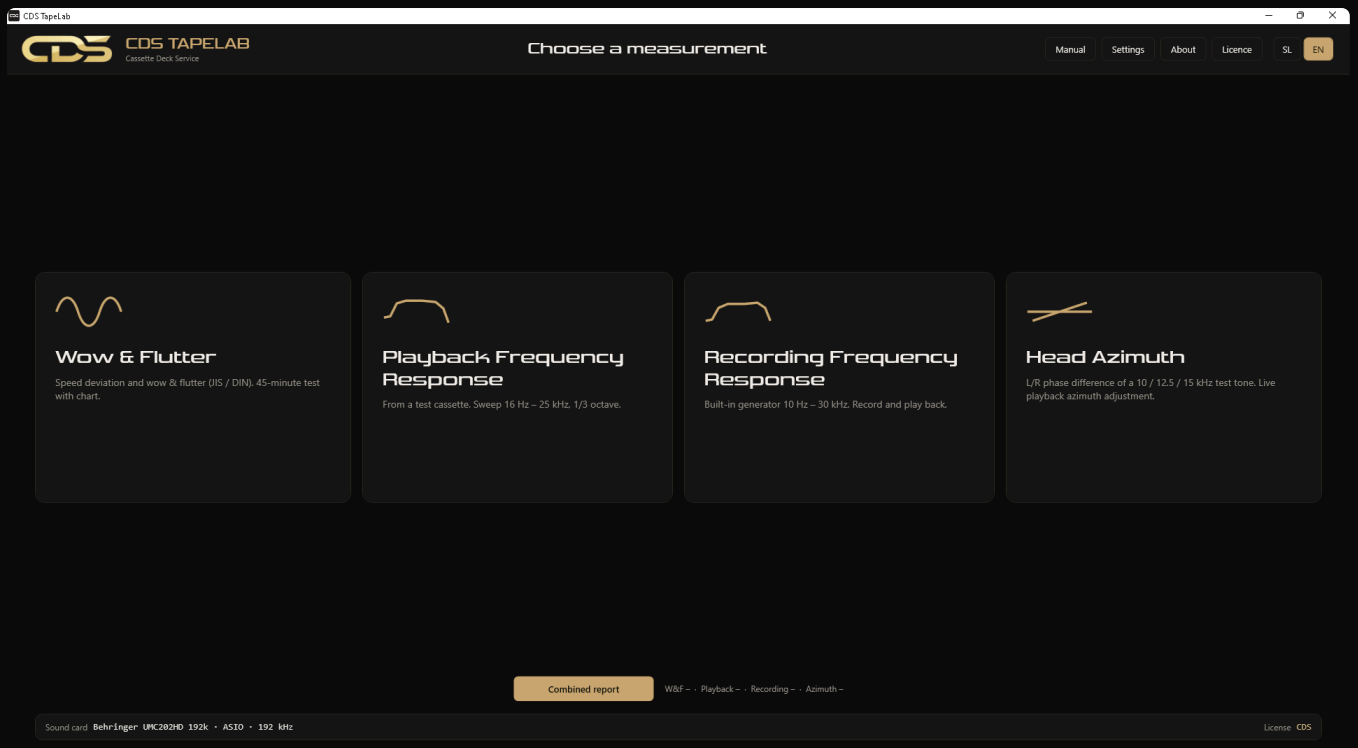
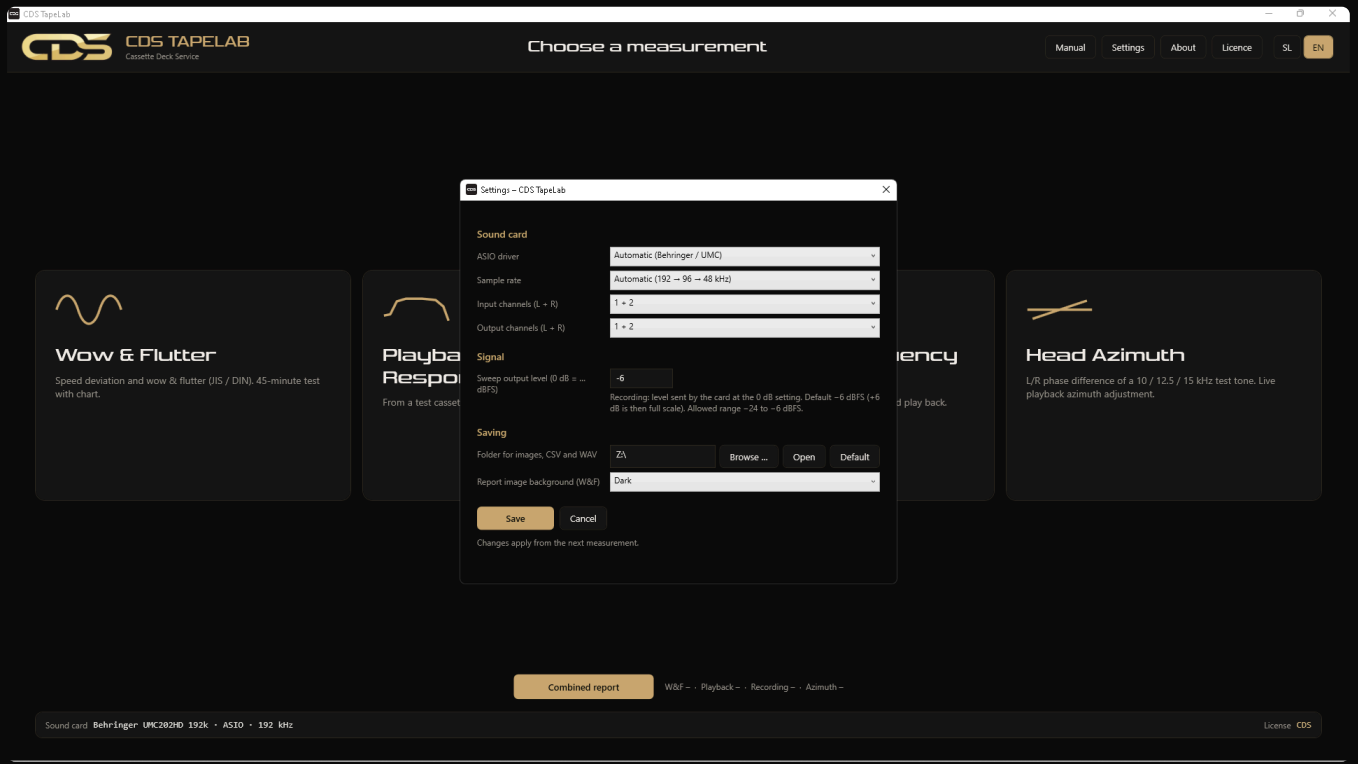


Figure: start screen; the Settings and About buttons are at the top right.

Settings opens a separate window. Changes take effect from the next measurement and are stored with **Save**; **Cancel** discards them.



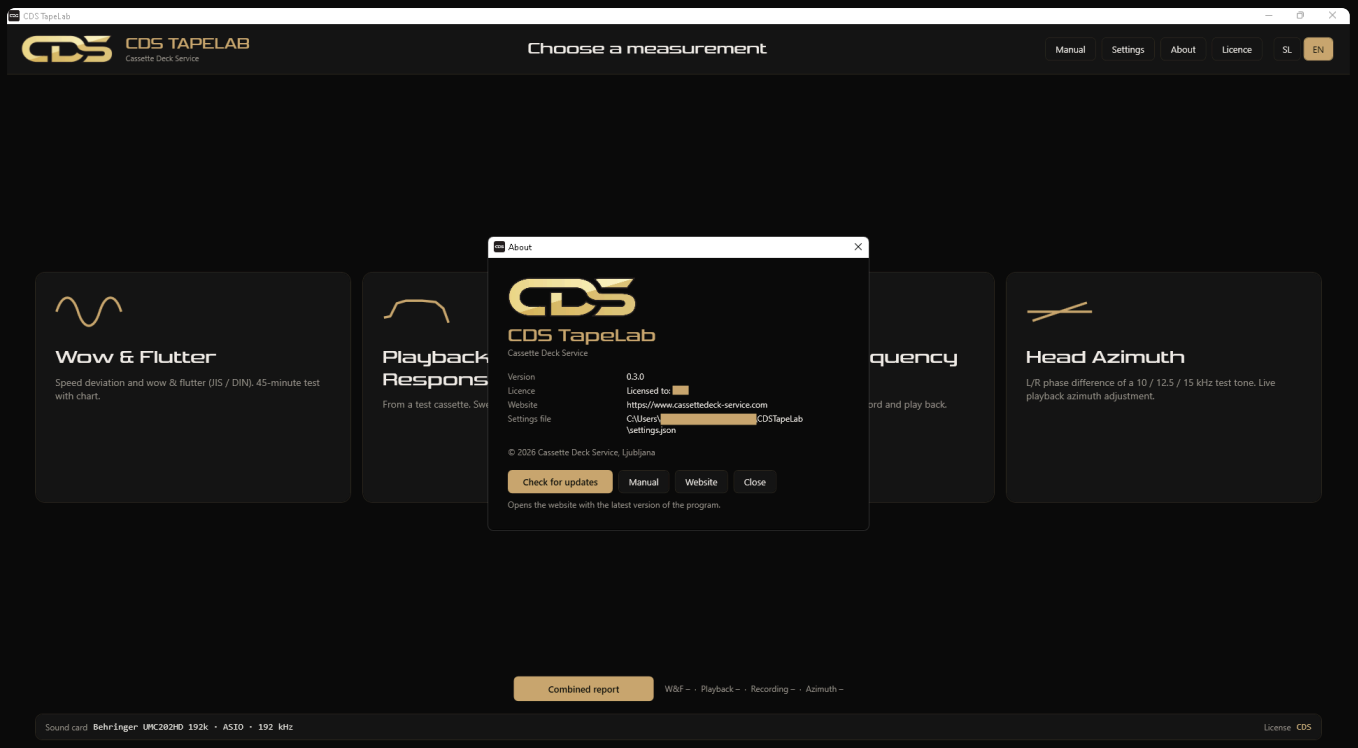
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Setting	Meaning
ASIO driver	The sound card. *Automatic (Behringer / UMC)* uses a Behringer / UMC card if one is installed, otherwise the first ASIO driver found; if you have several sound cards, pick the driver you want from the list.
Sample rate	*Automatic* tries 192, then 96 and finally 48 kHz. You can also choose one fixed value.
Input channels (L + R)	The pair of card inputs the cassette deck is connected to (e.g. 1 + 2).
Output channels (L + R)	The pair of card outputs that carries the generator signal in the Recording measurement.
Sweep output level	The level the card sends at the 0 dB setting in the Recording measurement. Default −6 dBFS (+6 dB is then full scale). Allowed range −24 to −6 dBFS.
Folder for images, CSV and WAV	Where the program saves exported files. **Browse …** picks a folder, **Open** shows it in Explorer, **Default** restores *Documents|CDS TapeLab*.
Report image background (W&F)	Background of the report image that the Screenshot button saves from the Report view of the Wow & Flutter measurement: Dark (default, as in the program) or White (for printing).

If the program cannot read the channel count of the selected driver, the window says *Could not read the channel count of the driver; channels 1–8 are shown*. In that case choose the channel pair manually.

About shows the program version, the licence status, the website and the location of the settings file (`%AppData%\CDSTapeLab\settings.json`).

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The **Check for updates** button opens the website with the latest version of the program; the **Website** button opens the home page. To update, simply install the new version over the old one.

3. Preparing for a measurement

1. Connect the sound card (e.g. the Behringer UMC202HD) to the computer via USB and switch it on.
2. Connect the cassette deck to the sound card as shown in the figure below. Input 1 of the card is the left channel, input 2 the right channel.
3. The ASIO driver of your sound card must be installed (the program finds it by itself). The sample rate is chosen automatically (192 kHz if the card supports it); see Settings.
4. Close other programs that use the sound card (Yoshimasa, media players) so that they do not occupy ASIO.
5. Start CDS TapeLab and choose a measurement on the start screen.

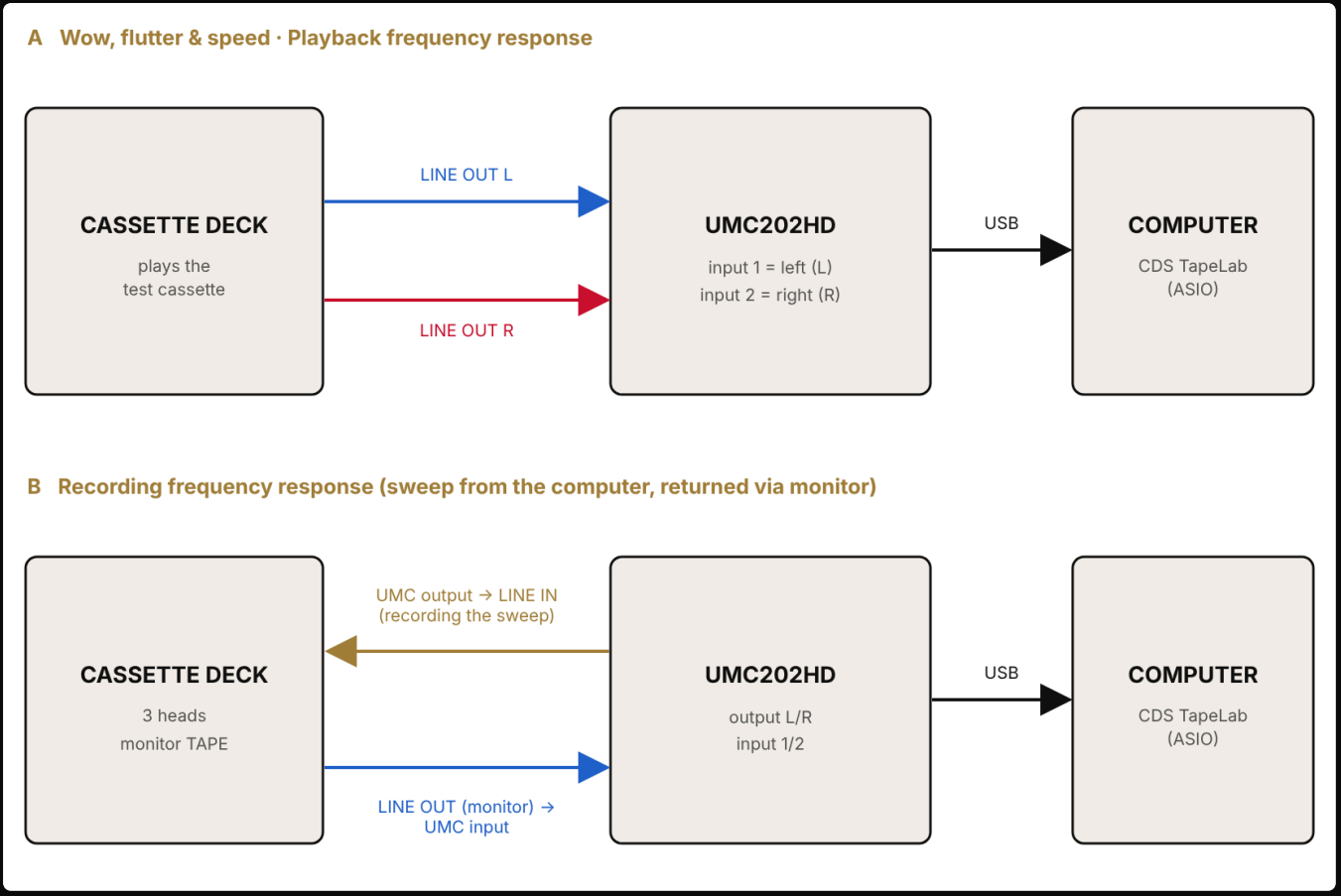


Figure: A = measurements from a cassette (wow & flutter, playback), B = recording measurement.

Cassettes you need

Measurement	Cassette
Wow, flutter & speed	test cassette with 3000 Hz (JIS) or 3150 Hz (DIN)
Playback frequency response	test cassette with a sweep
Recording frequency response	blank cassette, suitable for your deck
Head azimuth	test tape with an azimuth tone (10, 12.5 or 15 kHz)

The signal level during the measurement is shown to the right of the Start button (e.g. IN L −6.0 R −6.2 dBFS). It should not reach 0 dBFS.

4. Wow, flutter & speed

This measurement shows how evenly the tape runs: speed deviation in % and wow & flutter according to JIS (weighted RMS) or DIN (weighted peak). The test tone of 3000 Hz (JIS) or 3150 Hz (DIN) is measured from the cassette.

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Procedure

1. On the start screen click **Wow, flutter & speed**.
2. At the top choose the **Reference**: 3000 Hz or 3150 Hz, as on the test cassette.
3. Choose the **Input**: *Live (ASIO)* for a real measurement or *Demo* to try the program with simulated data. Choose the **Channel**: L, R or L+R.
4. Optionally choose the **mechanism/model** of the deck (Nakamichi Classic, Sankyo DD/BD/SD, ZX-9, Dragon; see "Choosing the mechanism and model" below). The program then marks in the spectrum which rotating part is the probable cause of a peak. "No profile" means that no profile is used.
5. Choose the **Duration** (15, 30, 45 or 60 min; default 45 min).
6. Play the test cassette and click ▶ **Start**. For the first few seconds the program shows "Settling…".
7. The measurement stops by itself when the tone disappears ("End of tape – measurement finished") or when the selected time has elapsed. You can also stop it manually with ■ **Stop**.



Figure: the Wow & Flutter window during a measurement (Sankyo SD mechanism selected, with the MAX marker).

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What the screen shows

Screen part	What you see
1. Speed deviation	a bar from −3 to +3 % and a number in Hz and % (e.g. 3000.9 Hz, +0.03 %)
2. Speed deviation in time domain	a chart of the deviation; window 1 s, 10 s, 60 s or All; "Periodicity" at the top (strongest frequency of the variation and its size)
3. Wow & flutter	a bar and a number in %. Mode: *Weighted RMS (JIS)*, *Weighted peak (DIN)* or *Unweighted*. Range: 0.05 %, 0.1 %, 0.5 % or 1.0 %
4. Wow & flutter in frequency domain	W&F versus frequency. Peaks (e.g. 1.38 Hz) show which part is varying. Window 4, 8 or 16 s: a larger window gives a more accurate frequency

If you select a model, the program also shows the factory **MAX** W&F for that model.

Choosing the mechanism and model

The mechanism buttons are in the top row ("Select mechanism/model"). The choice is optional, but with it the program shows the factory MAX W&F and marks the rotating parts in the spectrum. Choose the mechanism of your deck:

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Button	Models
Nakamichi Classic	ZX-7, RX-303, RX-505, 480Z, 481Z, 482Z, 580M, 581, 582, 582Z, 660, 660ZX, 670, 670ZX, 680, 680ZX, 681ZX, 682ZX, LX-3, LX-5, 700ZXE, 700ZXL, 1000ZXL
Sankyo DD	BX-300, CR-4, CR-5, CR-7, CR-40, ZX-5, MR-1
Sankyo BD	CR-3, Cassette Deck 1, Cassette Deck 1.5, DR-1, DR-2, DR-10
Sankyo SD	BX-1, BX-2, BX-100, BX-125, BX-150, MR-2, CR-1, CR-2, RX-202, Cassette Deck 2, DR-3, DR-8
Nakamichi ZX-9	ZX-9
Nakamichi Dragon	Dragon
No profile	the program uses no profile

The list of models is also shown as a tooltip when you point at a button. If the models of a mechanism have different factory maximums (Sankyo DD), a **Model** row appears after you click the button: choose your model. For the other mechanisms no model choice is needed. For Sankyo BD and SD the program shows only the factory MAX and does not mark peaks in the spectrum.

Report

The **Report** button (at the top right, next to "Live") shows the W&F&S chart for the whole measurement: speed, weighted-RMS W&F and weighted-peak W&F over time, with min, max and average values. The axes can be fixed (0–0.20 % and 2950–3050 Hz) or automatic. Save the report with the *Screenshot* or *Export CSV* button (licence required).

The *Import txt/csv file* button opens a record from other instruments (e.g. T100) so that you can show it in the same report.

5. Playback frequency response

This measurement shows how the deck reproduces the frequencies on a test cassette. The result is a chart with vertical bars in 1/3-octave bands from 16 Hz to 25 kHz (left channel blue, right channel red).

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Procedure

1. On the start screen click **Playback frequency response**.
2. Choose the **Mode**: *Sweep* (continuous sweep on the cassette) or *Steps* (individual tones).
3. Play the test cassette with the sweep and click ▶ **Start**. The program shows "Waiting for signal…", then "Measuring…"; the chart is drawn live.
4. When the sweep ends, the measurement stops by itself and automatically saves an image of the chart (licence required). You can also stop it manually with ■ **Stop**.

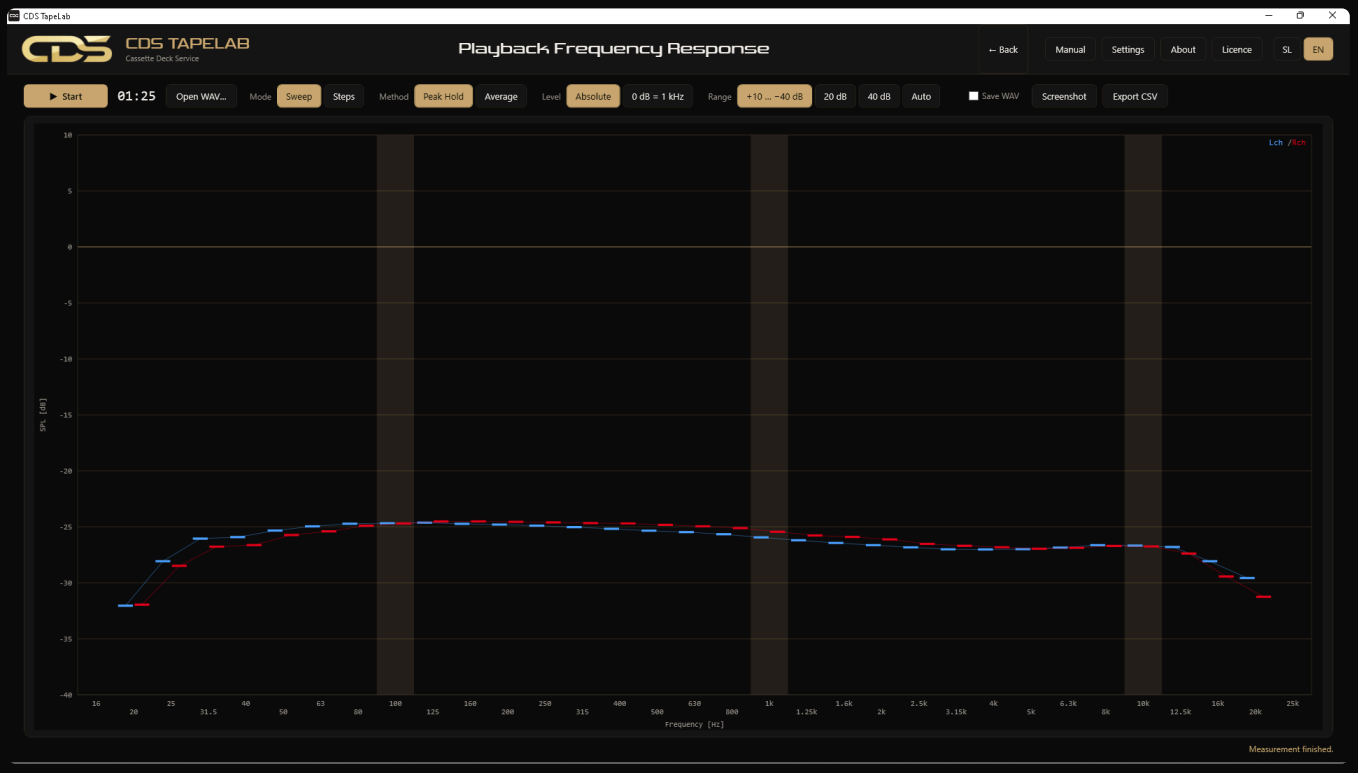


Figure: the Playback Frequency Response window after a finished measurement.

You can analyse an already recorded file with the **Open WAV…** button. With the tick at **Save WAV** the program also saves the raw recording during the measurement.

Settings above the chart

Setting	Choices	When
Method	*Peak Hold* (highest value in the band) or *Average*	Peak Hold for comparison with other instruments
Level	*Absolute* (dBFS, as measured) or *0 dB = 1 kHz* (chart normalised to 1 kHz)	Normalised when you only want to see the shape of the curve
Range	+10 … −40 dB, 20 dB, 40 dB or Auto	Auto if the curve is outside the range

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The chart shows the absolute level in dBFS.

You can save the measurement as an image (*Screenshot*) or as data (*Export CSV*; one column per channel and band). Both require a licence.

6. Recording frequency response

The program itself sends a sweep (from 10 Hz to 30 kHz), the deck records it, and the program immediately measures what comes back through the monitor (3rd head). Because the deck records and plays back at the same time, the cassette does not need to be rewound and the curve is drawn live.

Preparing the deck

1. Insert a blank cassette and connect the deck as shown in figure B in chapter 3.
2. Switch on the **monitor** (TAPE) so that you hear the signal from the third head, and prepare the deck for recording.
3. Set the recording level of the deck once and do not change it between measurements.

Procedure for one measurement

1. On the start screen click **Recording frequency response**.
2. Choose the **Sweep level**: −20, −10, 0 or +6 dB. This is the level that the program sends to the deck.
3. Choose the **Duration**: 10, 20, 30 or 60 s. A longer sweep gives a smoother and more accurate curve (60 s recommended).
4. Choose the **Scale**: *+10 … −40 dB* or *Auto*.
5. Start recording on the deck and click ▶ **Start**. The program sends the sweep and the chart is drawn live.
6. When the sweep is over (and a few seconds after it), the measurement stops by itself and saves an image of the chart (licence required).

Level in the program	Level at the card output
−20 dB	−26 dBFS
−10 dB	−16 dBFS
0 dB	−6 dBFS
+6 dB	0 dBFS

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All 4 measurements at once

The **All 4 measurements** button measures all four levels one after another (−20, −10, 0 and +6 dB) with a 60 s sweep:

1. Click **All 4 measurements**. The program deletes the old curves and measures the first level.
2. Between measurements there is a 5 s pause (the button shows "Now: …"; with it you can skip the pause).
3. After the last measurement the image is saved automatically.
4. If you click **Stop**, the sequence is interrupted; curves that have already been measured remain.

The whole sequence takes a little over 4½ minutes.

One chart for all measurements

All measurements are shown on **one chart**. The level is written next to each curve. The blue curve is the left channel, the red curve the right channel.

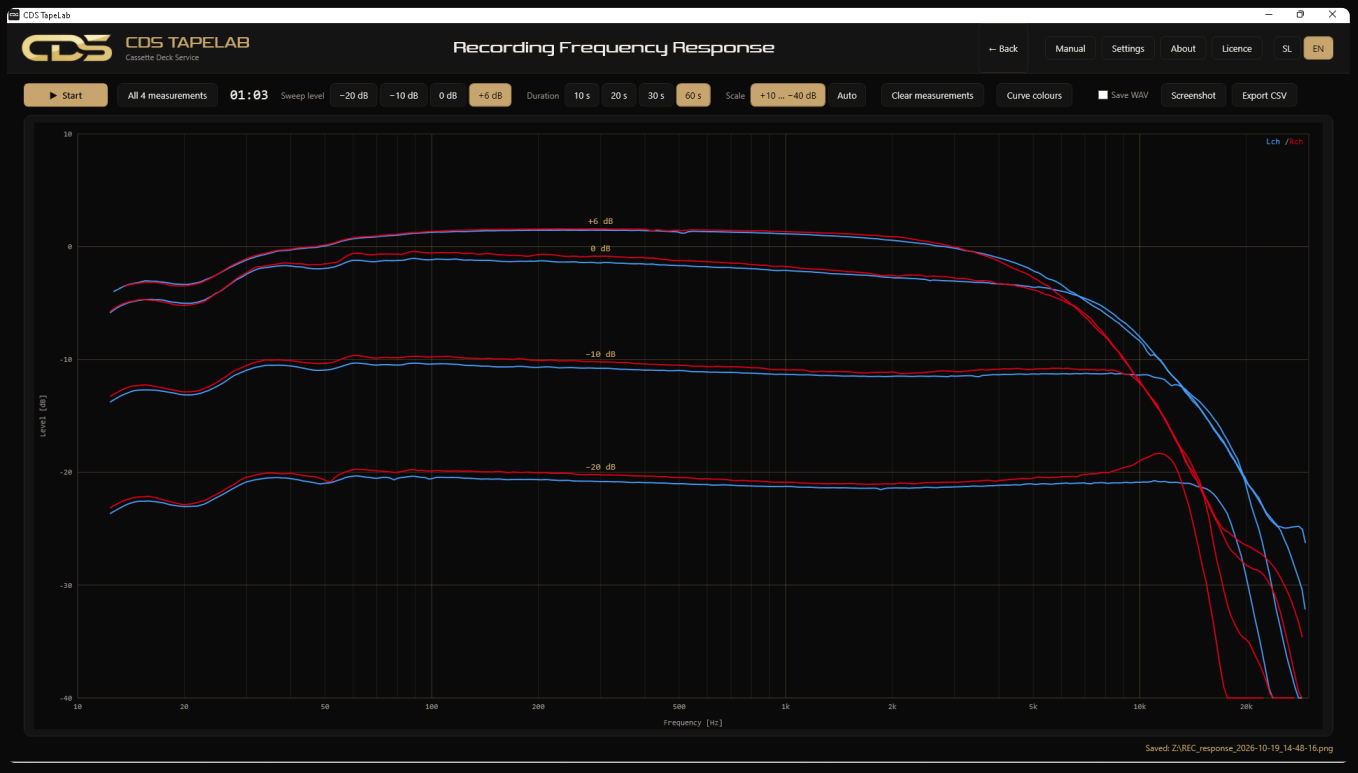


Figure: the Recording Frequency Response window after measuring at −20, −10, 0 and +6 dB.

Curve colours and comparison (pass thru)

Measurements at the same level are not overwritten. This way you can first record a **pass thru** (the signal as the program produces it, without a cassette), then measure with a cassette at the same level and see both together on the chart.

- The second measurement at the same level automatically gets different colours (cyan/magenta), the third green/orange.
- Change the colour of each curve with the **Curve colours** button: click the L or R field of a measurement, then a colour in the palette.
- **Clear measurements** deletes all curves and colours.
- The chosen colours are kept as long as the program is open.

7. Head azimuth

This measurement shows whether the playback head is perpendicular to the tape. With a wrong azimuth the left and right channels are shifted in phase and the high frequencies are quieter. The program measures the L − R phase difference of a high-frequency test tone from a test tape (10, 12.5 or 15 kHz) and shows it live, so you can adjust the azimuth screw on the head while watching. Only the playback azimuth is measured.

Procedure

1. On the start screen click **Head azimuth**.
2. Choose the **Tone**: *Auto* (the program finds the strongest tone between 5 and 18 kHz) or 10, 12.5 or 15 kHz, as on your test tape.
3. Choose the scale **Range** (±30°, ±90° or ±180°) and the **Green zone** (±2, ±5 or ±10 µs).
4. Play the part of the tape with the azimuth tone and click ▶ **Start**. It says “Waiting for tone …” and then “Measuring”.
5. Slowly turn the azimuth screw until the needle reaches 0 ° (inside the green zone) and the L+R level is at its maximum.
6. Click ■ **Stop**.

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Figure: the Head azimuth window during a measurement (needle, time delay, levels and the 30-s trace).

What the screen shows

Part of the screen	What you see
Phase difference	large number in degrees; 0 ° means both channels are in phase
Time delay	the same difference in µs (phase angle divided by 360 and the tone frequency)
Test tone	the tone frequency detected by the program
Needle	horizontal scale with a green zone; the needle should be inside the green
Trace	phase difference over the last 30 s; shows whether the adjustment is steady
Levels	L, R and L+R in dBFS, with the maximum L+R next to them (button *Reset max*)

The program warns if the input is clipping or if the channels differ by more than 3 dB. The input needs two channels (L and R). The green zone is an indicative guide; the real tolerance depends on the cassette deck.

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Picture, recording and report

The **Screenshot** button saves a picture of the window (PNG) and the ● **30 s MP4** button saves a 30-second video of the window; both need a licence. The files are named AZ_date_time and are saved in the export folder. The last valid reading and the 30-second trace are also kept for the combined report (chapter 8) when you stop the measurement or leave the window.

8. Exporting images and data

Export works only with a valid licence (chapter 2). In demo mode the program shows: "Demo: export is disabled. Enter a licence (Licence button) to export."

Button	What it does
Screenshot	saves an image of the chart (PNG, 1600 × 800 pixels)
Export CSV	saves the measured values to a table that you can open in Excel
Save WAV (tick)	also saves the raw recording during the measurement

The program saves all files in the folder **Documents \ CDS TapeLab**. The file name contains the type of measurement, the date and the time (e.g. `REC_response_2026-10-19_01-18-00.png`). After a playback or recording measurement the image is saved automatically; where it is stored is shown in the bottom line of the program ("Saved: …").

The CSV uses a semicolon as separator and a decimal point; it is written in UTF-8.

Combined report (PDF)

After one or more measurements (wow & flutter, playback, recording, azimuth) you can create a single report for your customer.

1. On the start screen click **Combined report**. The button is active once at least one measurement has been done in this session; next to it you can see which measurements are already included (✓).
2. Enter the **customer**, the **cassette deck (model)**, the **serial number**, the **date** and any **notes**.
3. Click **Create PDF**. The file is saved in the export folder (see Settings) and opens automatically.

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The report has two or three pages in the CDS style. The first contains the customer and deck details and the wow & flutter results (speed, W&F per JIS and DIN with min, max and average values, the factory maximum of the selected model and the chart). The second contains the Playback chart and the Recording chart. A third page is included only if you measured the azimuth in the session: it shows the phase difference, time delay, test tone, the needle with the green zone and the 30-second trace. Measurements that were not done in the session are marked as not done. The results are kept while the program is open; a new measurement of the same kind replaces the previous one. Creating the PDF requires a licence, like the other exports.

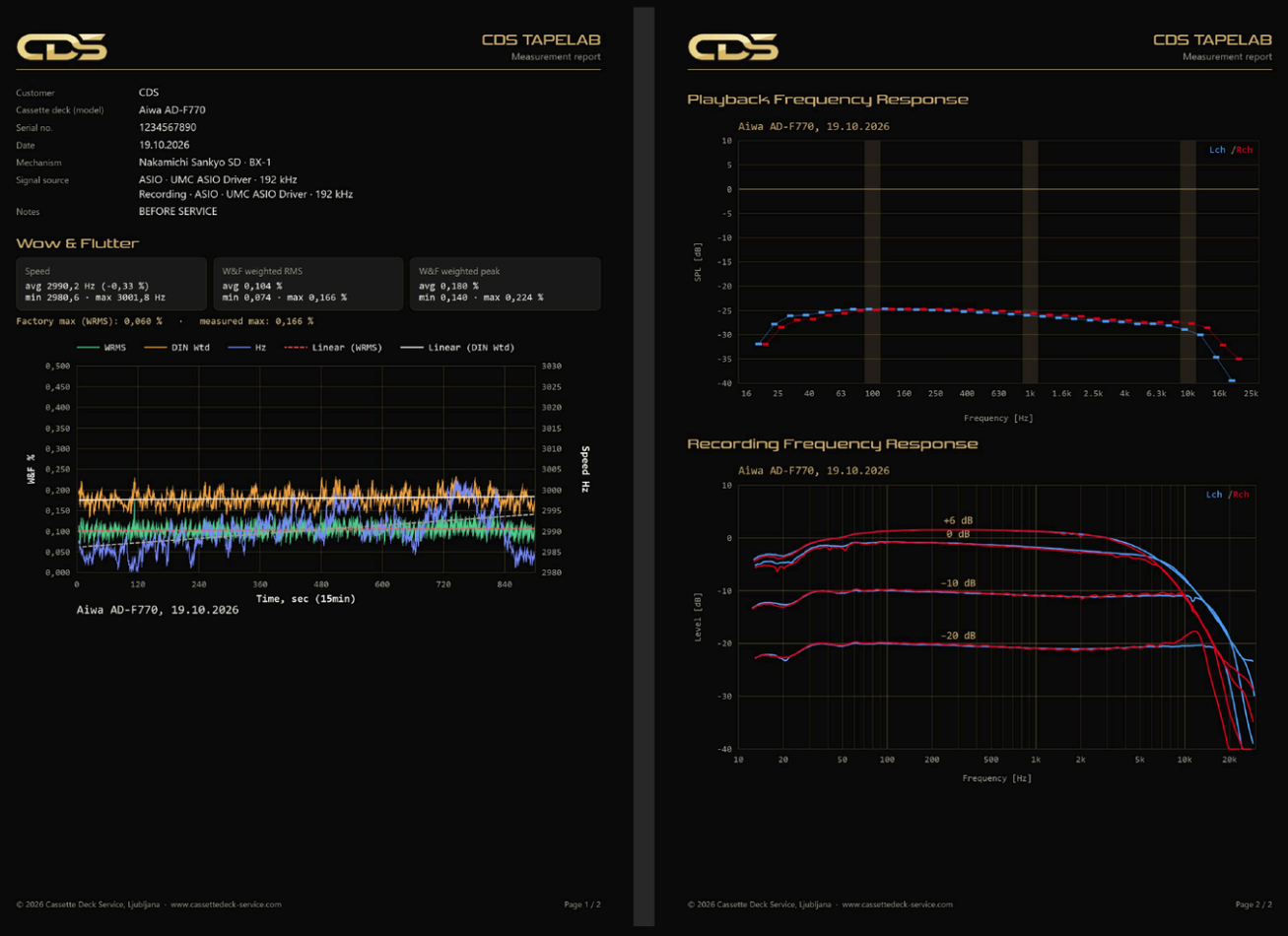


Figure: the two pages of the combined report (example).

9. Troubleshooting and tips

Problem	What to do
"No ASIO driver found."	Install the ASIO driver of your sound card (e.g. Behringer UMC ASIO), connect the card and restart the program

Problem	What to do
"ASIO error: …", especially when repeating a measurement	Close the measurement (Back button), open it again after a few seconds. If that does not help, close the program and start it again
"No signal (test tone not detected)"	Check the cables, the level on the deck and that the correct reference (3000 or 3150 Hz) is selected
"Not enough valid points."	The signal is too low or there is no sweep. Check the level (IN L/R below 0 dBFS) and the type of cassette
The chart falls apart at the end or the curve jumps	Use a longer sweep (30 or 60 s); above ~15 kHz the signal may be below the noise
Export does not work	Check the licence (Licence button). Without it export is disabled
Activation fails	Check the internet. A code works on one computer only

Tips for accurate measurements

- Do not run other heavy tasks on the computer during a measurement.
- For recording use a 60 s sweep for the smoothest curves.
- Before the real measurement record a **pass thru** as a reference and compare it with the cassette.
- Do not change the recording level on the deck between the four measurements.
- Measure W&F on the test tone of the cassette and not in the first seconds after start-up: settling takes a few seconds.

Support: www.cassettedeck-service.com

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